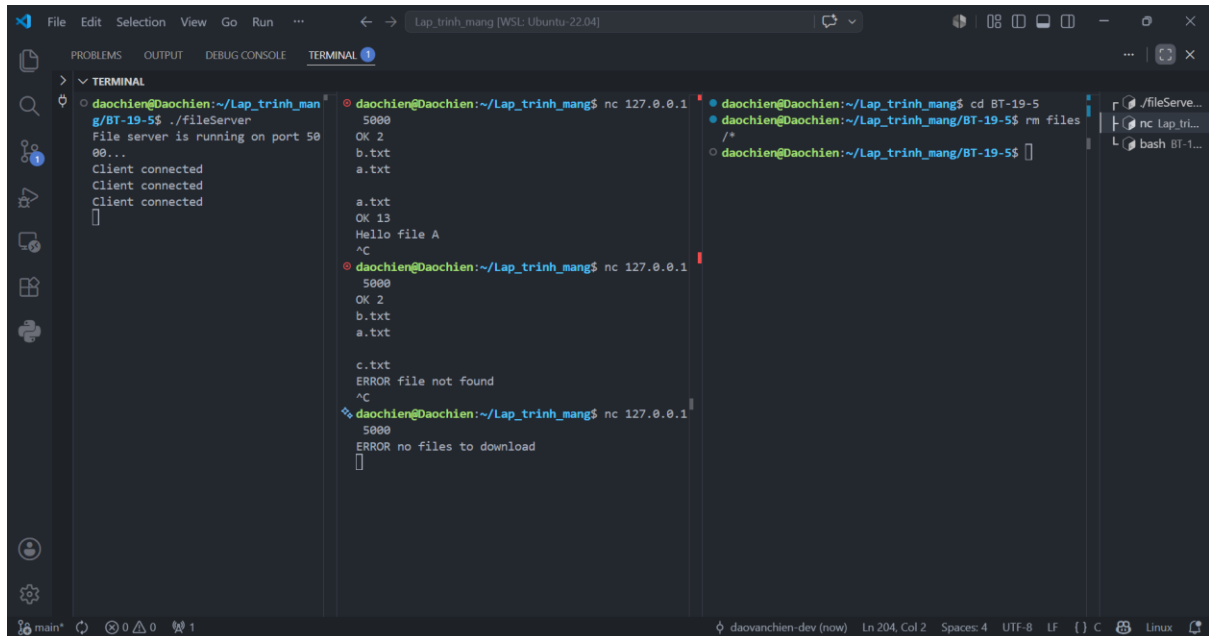


Đào Văn Chiến – 20225265

Bài tập về nhà 03

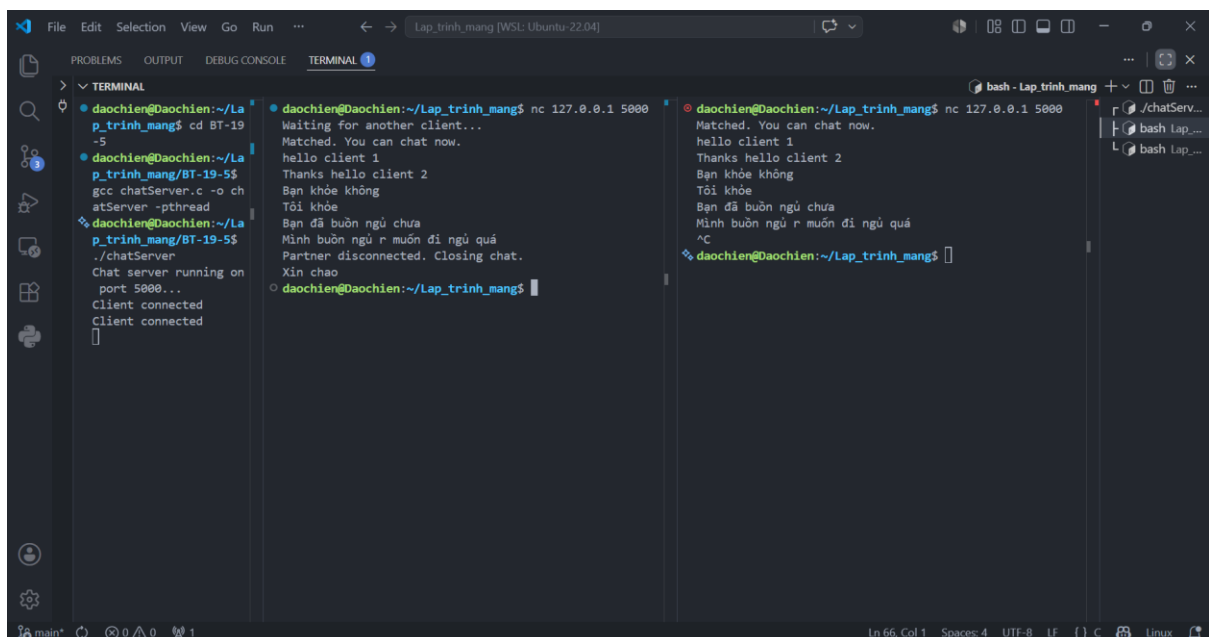
Bài1. Lập trình fileServer.c



```
daochien@Daochien:~/Lap_trinh_mang$ ./fileServer
File server is running on port 5000...
Client connected
Client connected
Client connected
daochien@Daochien:~/Lap_trinh_mang$ nc 127.0.0.1 5000
OK 2
b.txt
a.txt
a.txt
OK 13
Hello file A
^C
daochien@Daochien:~/Lap_trinh_mang$ nc 127.0.0.1 5000
OK 2
b.txt
a.txt
c.txt
ERROR file not found
^C
daochien@Daochien:~/Lap_trinh_mang$ nc 127.0.0.1 5000
ERROR no files to download
daochien@Daochien:~/Lap_trinh_mang$
```

- Khi client kết nối fileServer gửi tới số lượng và tên file
- Khi client gửi tên file để tải fileServer gửi kích thước file và file cho client
- Khi truy cập vào file không tồn tại fileServer sẽ gửi file not found
- Khi fileServer không có file nào sẽ gửi no files to down khi client kết nối

Bai2. Lập trình chatServer.c

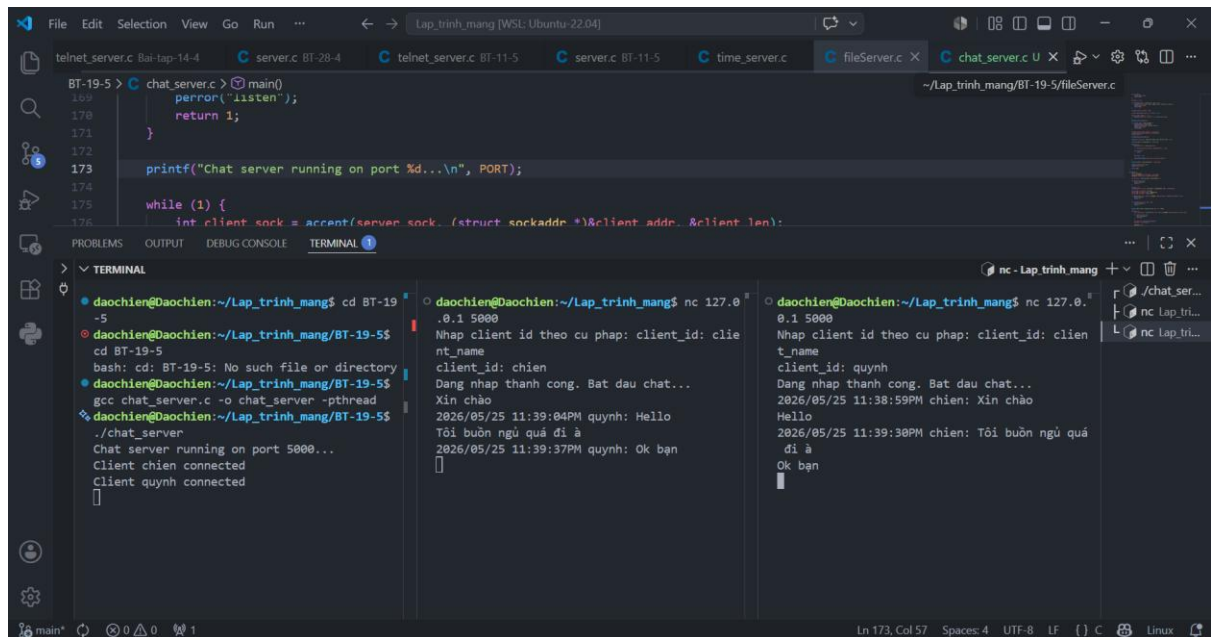


```
daochien@Daochien:~/Lap_trinh_mang$ ./chatServer
Chat server running on port 5000...
Client connected
Client connected
daochien@Daochien:~/Lap_trinh_mang$ nc 127.0.0.1 5000
Waiting for another client...
Matched. You can chat now.
hello client 1
Thanks hello client 2
Bạn khỏe không
Tôi khỏe
Bạn đã buồn ngủ chưa
Mình buồn ngủ r muốn đi ngủ quá
Partner disconnected. Closing chat.
Xin chào
daochien@Daochien:~/Lap_trinh_mang$
```

- Khi 2 client kết nối tới server sẽ được kết nối chat với nhau
- Một client thoát thì client còn lại cũng sẽ đóng kết nối

Bài tập về nhà 04.

Bai1. Lập trình chat_server dùng kiến thức multithread



The screenshot shows the VS Code editor with the file `chat_server.c` open. The code is as follows:

```
169 > chat_server.c > main()
170     perror("listen");
171     return 1;
172 }
173 printf("Chat server running on port %d...\n", PORT);
174
175 while (1) {
176     int client_sock = accept(server_sock, (struct sockaddr *)&client_addr, &client_len);
```

The terminal output shows the server running on port 5000 and handling two clients:

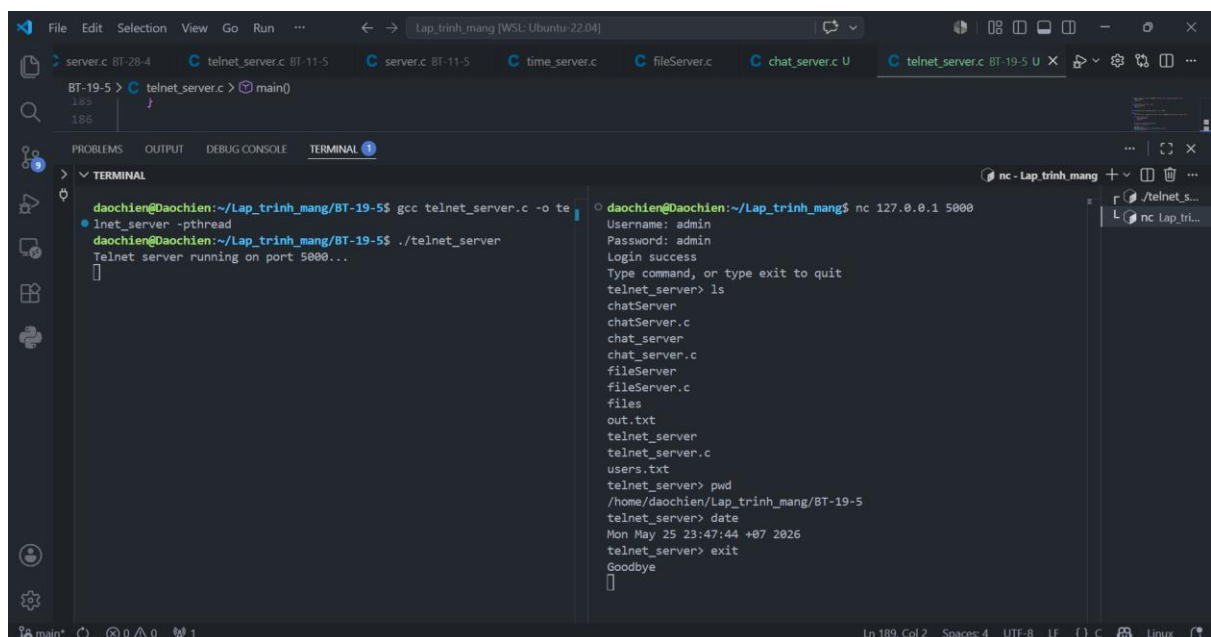
```
daochien@Daochien:~/Lap_trinh_mang$ cd BT-19
daochien@Daochien:~/Lap_trinh_mang/BT-19$ cd BT-19-5
bash: cd: BT-19-5: No such file or directory
daochien@Daochien:~/Lap_trinh_mang/BT-19-5$ gcc chat_server.c -o chat_server -pthread
daochien@Daochien:~/Lap_trinh_mang/BT-19-5$ ./chat_server
Chat server running on port 5000...
Client chien connected
Client quynh connected
```

Two terminal windows show client interactions:

```
daochien@Daochien:~/Lap_trinh_mang$ nc 127.0.0.1 5000
.0.1 5000
Nhap client id theo cu phap: client_id: client_name
client_id: chien
Dang nhap thanh cong. Bat dau chat...
Xin chào
2026/05/25 11:39:04PM quynh: Hello
Tôi buồn ngủ quá đi à
2026/05/25 11:39:37PM quynh: Ok bạn
```

```
daochien@Daochien:~/Lap_trinh_mang$ nc 127.0.0.1 5000
.0.1 5000
Nhap client id theo cu phap: client_id: client_name
client_id: quynh
Dang nhap thanh cong. Bat dau chat...
2026/05/25 11:38:59PM chien: Xin chào
Hello
2026/05/25 11:39:30PM chien: Tôi buồn ngủ quá đi à
Ok bạn
```

Bai2 . Lập trình telnet_server dùng kiến thức multithread



The screenshot shows the VS Code editor with the file `telnet_server.c` open. The code is as follows:

```
185 > telnet_server.c > main()
186 }
```

The terminal output shows the server running on port 5000 and handling a client connection:

```
daochien@Daochien:~/Lap_trinh_mang/BT-19-5$ gcc telnet_server.c -o telnet_server -pthread
daochien@Daochien:~/Lap_trinh_mang/BT-19-5$ ./telnet_server
Telnet server running on port 5000...
```

A terminal window shows a client interaction:

```
daochien@Daochien:~/Lap_trinh_mang$ nc 127.0.0.1 5000
Username: admin
Password: admin
Login success
Type command, or type exit to quit
telnet_server> ls
chatServer
chatServer.c
chat_server
chat_server.c
fileServer
fileServer.c
files
out.txt
telnet_server
telnet_server.c
users.txt
telnet_server> pwd
/home/daochien/Lap_trinh_mang/BT-19-5
telnet_server> date
Mon May 25 23:47:44 +07 2026
telnet_server> exit
Goodbye
```

Bai3. Lập trình _server dùng kiến thức multithread

```
daochien@Daochien:~/Lap_trinh_mang/BT-19-5$ gcc t
timeServer.c -o timeServer -pthread
cc1: fatal error: timeServer.c: No such file or d
irectory
compilation terminated.
daochien@Daochien:~/Lap_trinh_mang/BT-19-5$ gcc t
ime_server.c -o time_server -pthread
daochien@Daochien:~/Lap_trinh_mang/BT-19-5$ ./tim
e_server.c
bash: ./time_server.c: Permission denied
daochien@Daochien:~/Lap_trinh_mang/BT-19-5$ ./tim
e_server
Time server is running on port 5000...
New client connected
[]

nc - Lap_trinh_mang 127.0.0.1 5000
Time Server Ready
Command: GET_TIME [format]
Formats: dd/mm/yyyy, dd/mm/yy, mm/dd/yyyy, mm/dd/
yy
ERROR Invalid command
GET_TIME dd/mm/yyyy
OK 25/05/2026
GET_TIME mm/dd/yyyy
OK 05/25/2026
GET_TIME mm/dd/yy
OK 05/25/26
GET_TIME yyyy/mm/dd
ERROR Invalid format
```

Bai4. Lập trình http_server dùng kiến thức multithread

```
daochien@Daochien:~/Lap_trinh_mang/BT-19-5$ gcc http_server.c -o http_server -pthread
daochien@Daochien:~/Lap_trinh_mang/BT-19-5$ ./http_server
Server is running on port 8080...
New client connected: 4
Client 4 sent:
hello server

New client connected: 5
Client 5 sent:
xin chao http_server
[]

nc - Lap_trinh_mang 127.0.0.1 8080
hello server
HTTP/1.1 200 OK
Content-Type: text/html

<html><body><h1>Xin chao cac ban</h1></body>
</html>

daochien@Daochien:~/Lap_trinh_mang$ nc 127.0.0.1 8080
xin chao http_server
HTTP/1.1 200 OK
Content-Type: text/html

<html><body><h1>Xin chao cac ban</h1></body><
/html>
```